Infrastructure As Code (IAC)

Quand les sysadmin se mettent à coder ...

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Cours repris de Pr. Olivier Barais

Continuous delivery;)

A long time ago, in a data center far, far away, an ancient …

IAC

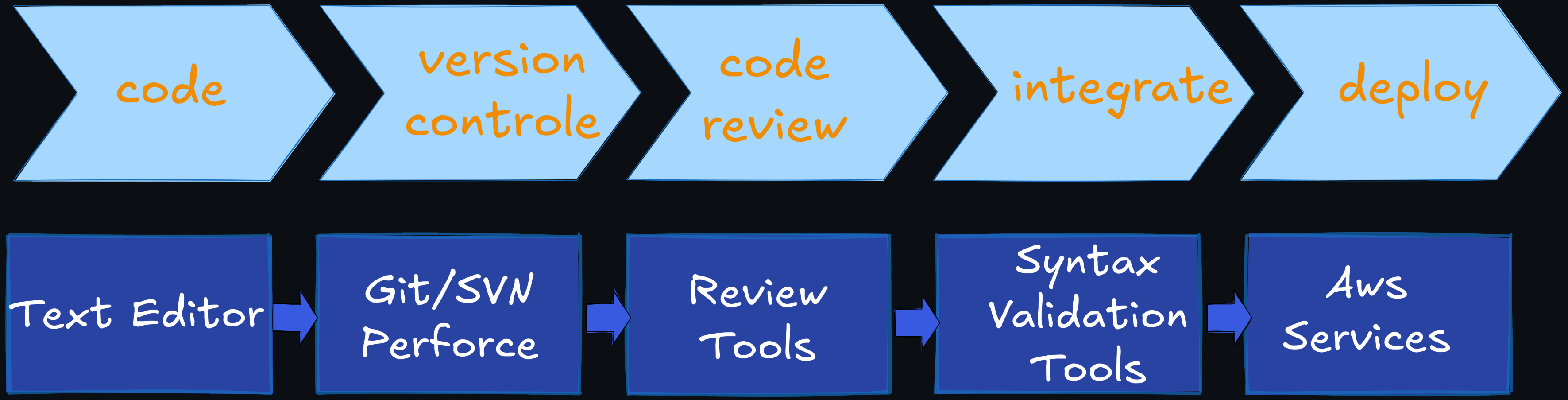
L'IAC fait appel à un langage descriptif pour coder des processus de déploiement Plus polyvalents et adaptatif que des scripts

Des descripteurs pour piloter des outils de configuration et de déploiement

- Avec ces outils il est possible de :
- Provisionner et déployer – D’avoir l’état d’une infra (suivant l’outil) – De gérer le code en le versionnant – De mettre en place des tests sur les déploiements – Créer des briques réutilisables

Quand les Sysadmins découvrent le génie logiciel ;)

IAC Workflow



Un langage descriptif

YAML

- acronyme récursif de YAML Ain’t Markup Language
- est un format de représentation de données par sérialisation – Unicode.

Wikipedia

Yaml

Basic YAML Syntax Rules

Example:

```yml
Student-Id: 1234
Name: John Doe

Address:
  Street: 123 Main St
  City: New York
  State: NY
  Zip: 10001

Phone-Number:
  Home:  555-1234
```

Les outils

HashiCorp Terraform

SALTSTACK

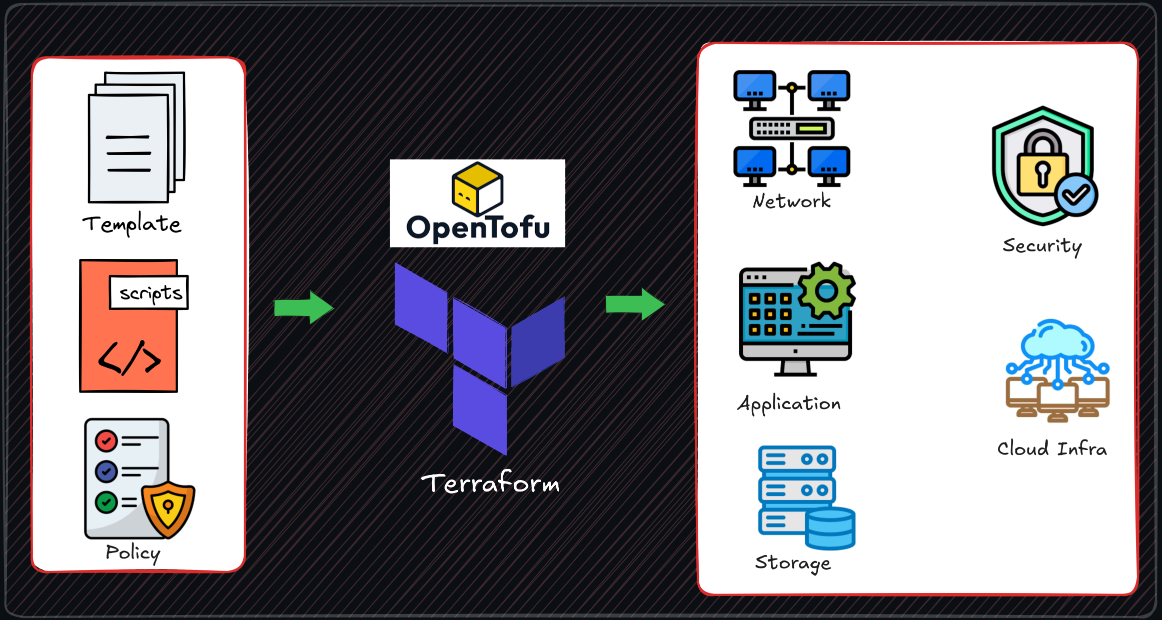
CHEF

puppet

CloudFormation

ANSIBLE

Un exemple : Terraform/OpenTofu



Un exemple : Terraform/OpenTofu

- Terraform est un outils open source qui permet de:
  - **définir** et **déployer** une infrastructure chez les différents fournisseurs cloud (e.g. AWS, Azure, Google Cloud, DigitalOcean, OVH,etc)
  - tout cela à partir d’un langage déclaratif

Il est possible d’utiliser Terraform avec une liste de providers importante :

– https://www.terraform.io/docs/providers/index.html

Chacun étant documenté :

– https://www.terraform.io/docs/providers/aws/index.html

Exemple :

Configure the aws provider

```tf
provider "aws" {
  version = "~> 2.0"
  region = "us-east-1"
}
```

Exemple Terraform avec OVH

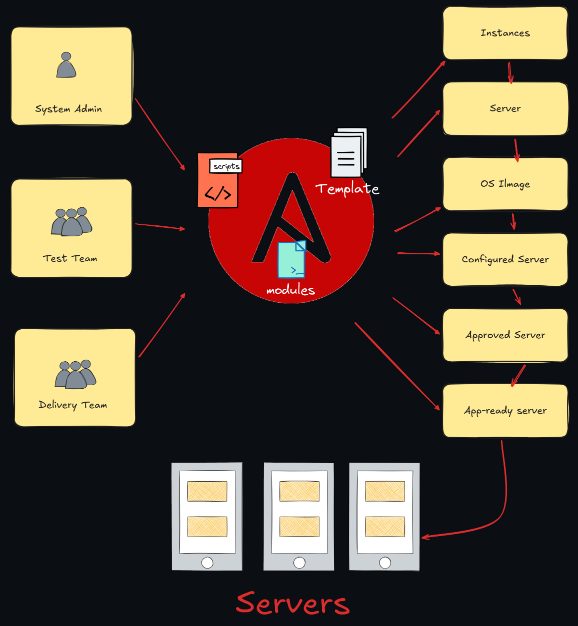
Terraform 0.13 and later:

```tf
terraform {
  required_providers {
    ovh = {
      source = "ovh/ovh"
    }
  }
}

provider "ovh" {
  endpoint      = "ovh-eu"
  client_id     = "xxxxxxxxx"
  client_secret = "yyyyyyyyy"
}
```

https://registry.terraform.io/providers/ovh/ovh/latest/docs

Exemple 2 : Ansible

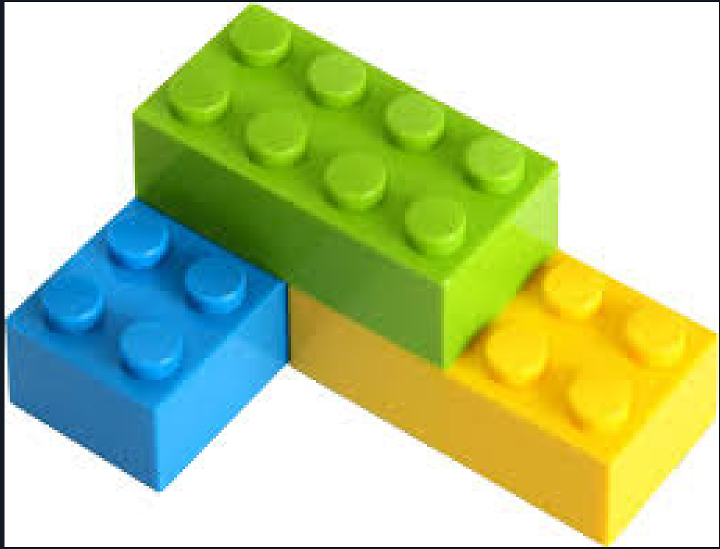


Exemple de playbook

```yaml
---
- hosts: web
  become: yes
  tasks:
    - name: Install Apache2
      apt:
        update_cache: yes
        name: apache2
        state: present
    - name: Copy file
      copy:
        src: "{{ item.src }}"
        dest: "{{ item.dest }}"
        owner: "{{ item.owner }}"
        group: "{{ item.group }}"
      with_items:
```

```yaml
        - src: "files/conf/apache/security.conf"
          dest: "/etc/apache2/conf-available/
          security.conf"
          owner: root
          group: root
        - src: "files/conf/apache/ports.conf"
          dest: "/etc/apache2/ports.conf"
          owner: root
          group: root
        - src: "files/web/index.html"
          dest: "/var/www/html/index.html"
          owner: www-data
          group: www-data
    - name: Make sure Apache is started
      service:
        name: apache2
```

Une architecture extensible



Un besoin ? Un module ou presque …

https://docs.ansible.com/ansible/latest/modules/list_of_all_modules.html

Si pas de module … le développer en python ..

Les facts ansible

`ansible web -m setup`

```
web01 | SUCCESS => {
  "ansible_facts": {
    "ansible_hostname": "web01",
    "ansible_fqdn": "web01.lab.local",
    "ansible_domain": "lab.local",
    "ansible_os_family": "Debian",
    "ansible_distribution_version": "22.04",
    "ansible_kernel": "5.15.0-91-generic",
    "ansible_architecture": "x86_64",
    "ansible_default_ipv4": {
      "address": "192.168.56.21",
      "gateway": "192.168.56.1",
      "interface": "eth0"
    },
```

```
    "ansible_processor": [
      "0",
      "GenuineIntel",
      "Intel(R) Core(TM) i7-8650U CPU @ 1.90GHz"
    ],
    "ansible_processor_vcpus": 2,
    "ansible_memtotal_mb": 2048,
    "ansible_env": {
      "HOME": "/root",
      "SHELL": "/bin/bash",
    }
  },
  "changed": false
}
```

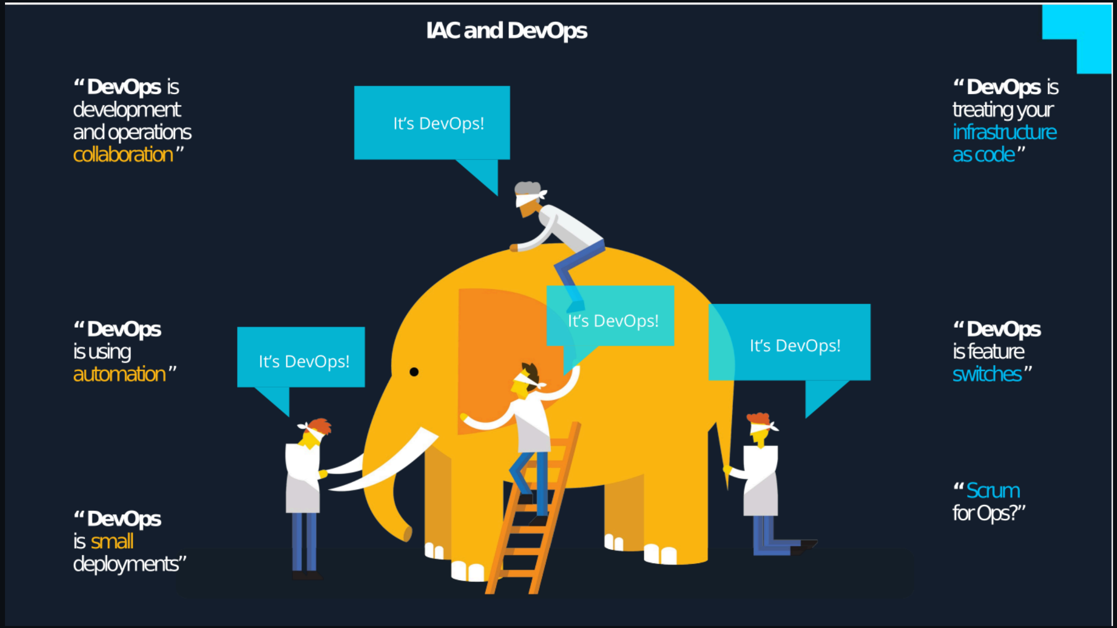
Ansible : Ninja un langage de template

```jinja2
<html>
  <head>
    <meta charset="UTF-8">
    <title>Server {{ ansible_hostname }}</title>
  </head>
  <body>
    <h1>Hello World!</h1>
    <h2>Le serveur Web {{ ansible_hostname |
    capitalize }} fonctionne !! </h2>
    <p>Cette page à été créée le
    {{ ansible_date_time.date }}.</p>
    {% if ansible_eth0.active == True %}
    <p>eth0 address {{ ansible_eth0.ipv4.address }}.</p>
    {% endif %}
    <p>Ce serveur fonctionne sur un système Linux de type
    {{ ansible_os_family}}, version
```

```
    {{ ansible_lsb.codename }}. </p>
    <p>Cette machine peut avoir plusieurs adresses ip :</
    p>
    <ul>
    {% for address in ansible_all_ipv4_addresses %}
    <li>{{ address }}</li>
    {% endfor %}
    </ul>
  </body>
</html>
```

Conclusion

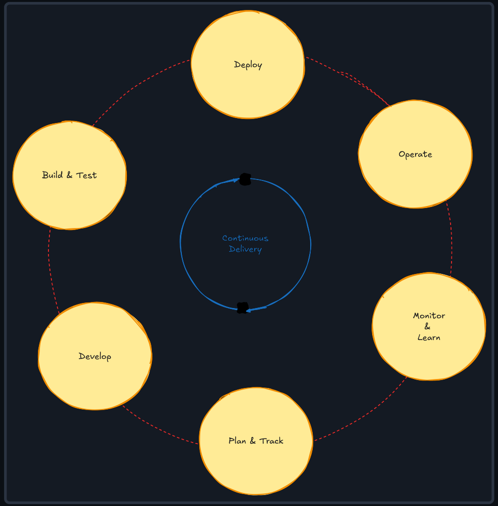
IAC and DevOPs


IAC and DevOps
"DevOps is development and operations collaboration"
It's DevOps!
"DevOps is treating your infrastructure as code"
"DevOps is using automation"
It's DevOps!
It's DevOps!
It's DevOps!
"DevOps is feature switches"
"DevOps is small deployments"
"Scrum for Ops?"

WHAT IS DEVOPTS

People. Process. Products.

> “DevOps is the union of **people**, **process**, and **products** to enable continuous delivery of **value** to your end use.”
>
> — Donovan Brown (Microsoft)



DevOps Best Practices

Version everything
Automate everything
Tokenize configurations
Use one-click deployments
Deploy the same way to every environment
Have always a rollback mechanism in place
Build only once
Lock down the environments
...